

Q3.1

Head = [1, 2, 3, 4, 5]
 ↑ ↑
 Head Tail.

Isko aa reverse karna hai.

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$[5 \quad 4 \quad 3 \quad 2 \quad 1]$
 $\uparrow \qquad \qquad \qquad \uparrow$
 Head \qquad \qquad \qquad Tail

To reverse Kaone Ke Gye.

Head = [1, 2, 3, 4, 5].

White Head = 1 Head

```
if (swap (head, tail)) {
    head = tail;
    tail = head;
}
```

proceso de -

curr Node

nest no. 1

Null $\leftarrow$ 1 $\xrightarrow{\text{X}}$ 2 $\xrightarrow{\text{X}}$ 3 $\xrightarrow{\text{X}}$ 4 $\xrightarrow{\text{X}}$ 5 $\xrightarrow{\text{X}}$ Null.

~~fixed.~~

↑
Head.

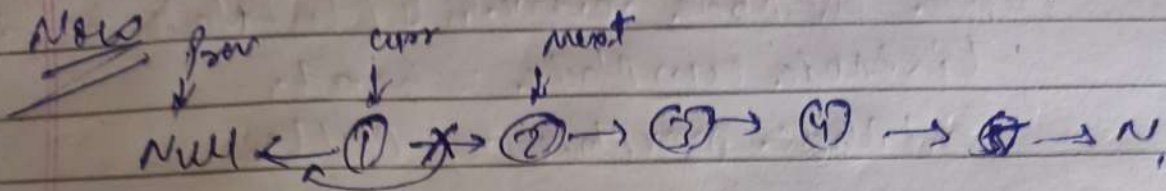
shese pehle km:-

ek new Node create karenge jo point
karega head ko.

* ek prevNode create karenge jo point karega null ko.

4. ek ~~curr~~ Node nextNode create karenge jo
point karaga currNode.next ab to-

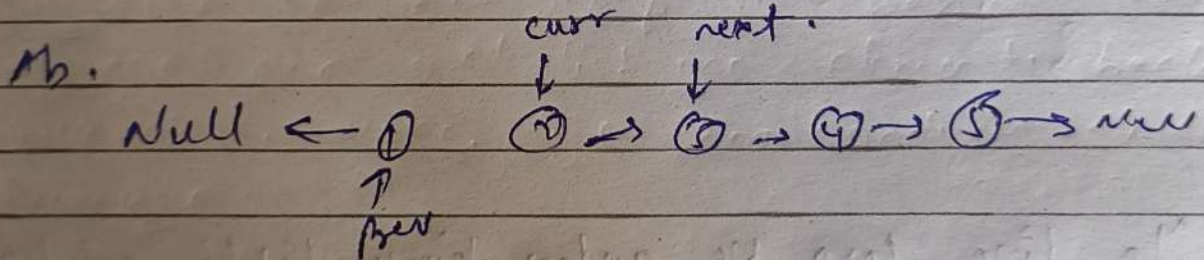
1. $currNode = Head$
 $prevNode = null$
 $NextNode = currNode.next$



$curr.next = prev$ i.e. curr ke next me prev. ko point karana honge

$prev = curr$ i.e. prev. ko update karenge next ke saath.

$curr = next$ i.e. curr ko update karenge next se.



Ab phir se curr 3 ko point kar raha hai, to uske jise hm phir se upper wala concept use karunga. i.e. ek loop laga denge.

while (currNode != null) {

1.

return prevNode.

Code

```
class soln {  
    public ListNode reverseList(ListNode head) {  
        'ListNode currNode = head;  
        ListNode prevNode = null;  
  
        while (currNode != null) {  
            'ListNode nextNode = currNode.next;  
            'currNode.next = prevNode;  
            prevNode = currNode;  
            currNode = nextNode;  
        }  
        return prevNode;  
    }  
}
```

→ Ye line loop ke andar bhi likhi gayi hai
↳ because! har iteration me currNode
change ho raha hai, isliye uska next
bhi har bar naya hoga.